

PROVISIONAL PATENT APPLICATION

Under 35 U.S.C. § 111(b)

Contact Resistant Breathing Trainer

Inventor(s): Christopher Sawicki, Michael Alexiadis, Christopher Miranda Soto

Address: 319 College Ave, Ithaca NY, 14850

Citizenship: United States

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FIG. 1

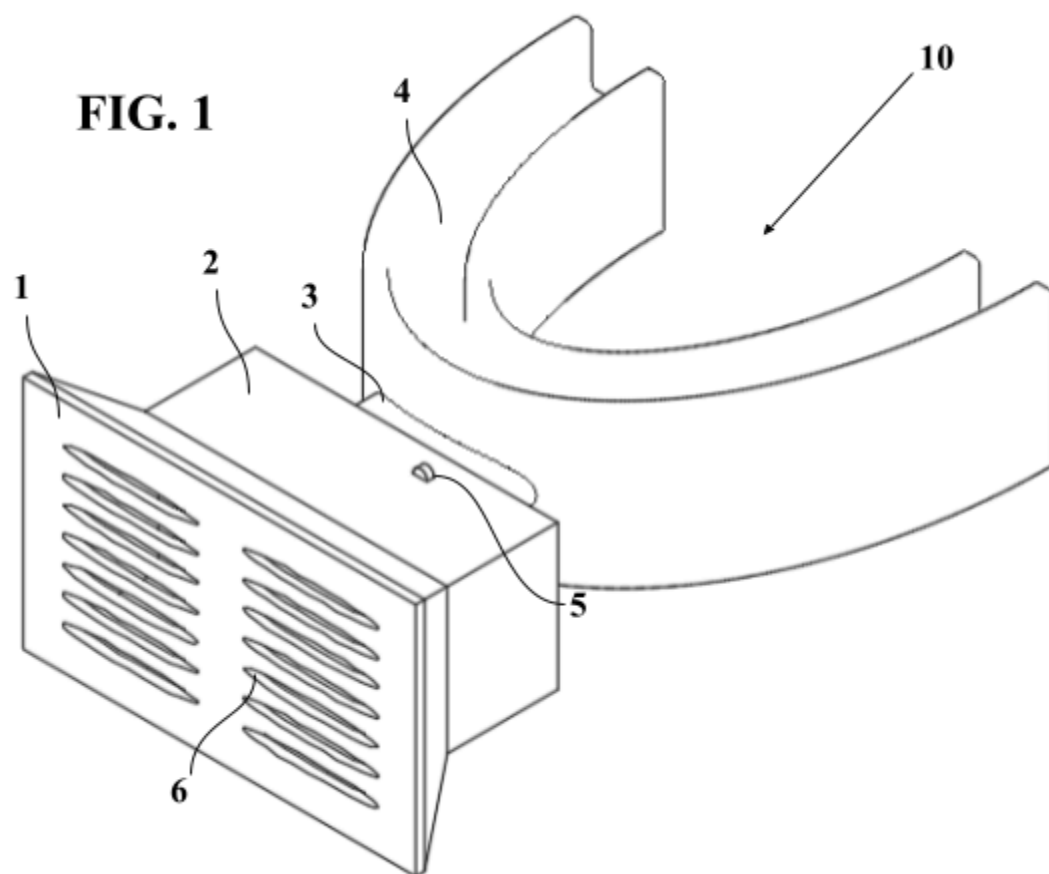


FIG. 2

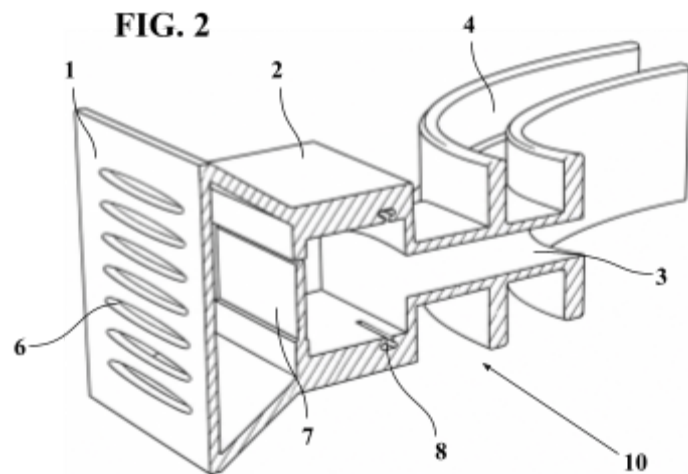
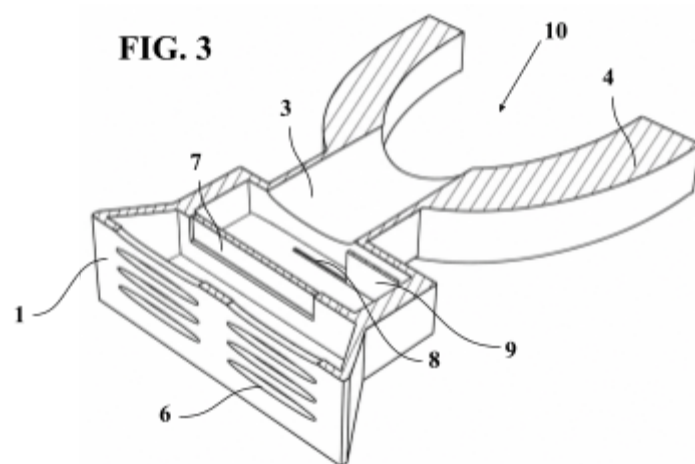


FIG. 3



TITLE OF THE INVENTION

Compact and Collision Resistant Mouth-Held Respiratory Muscle Trainer

FIELD OF THE INVENTION

The present invention relates generally to respiratory training devices, specifically to hand or mouth held trainers that challenge respiratory muscles with a variety of resistance levels.

BACKGROUND OF THE INVENTION

Respiratory muscle training is an established method for improving athletic performance and cardiovascular health. The inspiratory and expiratory muscles, including the diaphragm and intercostal muscles, respond to progressive resistance loading in much the same way that skeletal muscles respond to weight training. Studies have shown that consistent respiratory muscle training can improve endurance, reduce perceived breathlessness during exertion, and accelerate recovery in athletes across a wide range of sports.

Existing respiratory training devices fall into two broad categories. The first category includes handheld threshold devices, in which the user breathes through a spring-loaded or adjustable valve that requires a minimum pressure effort to open. These devices are effective for isolated respiratory training sessions but cannot be used during active athletic training, as they require the user to hold the device with one hand, disrupting normal movement and making them entirely unsuitable for contact sports or dynamic exercise. The second category includes wearable or mask-based respiratory trainers that fit over the nose and mouth. While these devices free the user's hands, they are bulky, impede vision, restrict airflow to a degree that may interfere with high-intensity training, and present a serious safety hazard in contact sports, where an impact to a rigid or semi-rigid face-mounted device can cause injury to the athlete or an opponent.

Neither category of existing device adequately addresses the needs of athletes who wish to train their respiratory muscles simultaneously with their normal training routines. A need therefore exists for a respiratory training device that (a) is held in the mouth without requiring the user's hands, (b) is constructed of compliant, collision-resistant materials suitable for contact sports, (c) provides manually adjustable resistance levels that can be changed during use, and (d) provides performance feedback to the user through an integrated sensor and application interface, without imposing a form factor that interferes with athletic performance.

SUMMARY OF THE INVENTION

Disclosed is a compact, mouth-held respiratory muscle training device comprising a mouthguard, a sensor chamber, and a protective chamber. The device is preferably formed from injection-molded silicone or similarly compliant materials, allowing an athlete to wear the device during contact sports and dynamic training without risk of injury from collision.

The device presents breathing resistance through two complementary mechanisms. A threshold flap disposed at the boundary between the protective chamber and the sensor chamber requires the athlete to

generate a minimum inspiratory or expiratory pressure to deflect the flap and permit airflow, thereby loading the respiratory muscles with every breath. In a preferred embodiment, the threshold flap is dimensioned to deflect at approximately 2.45 kPa (25 cmH₂O), corresponding to approximately 20% of the average athlete's Maximum Inspiratory Pressure. An adjustable resistance wall, controlled by an externally accessible resistance adjustment wheel, can be positioned to cover a variable portion of the airway inlet, providing additional resistance on top of the threshold mechanism and allowing the athlete to progressively increase training load.

The protective chamber surrounds the threshold flap and is vented through a pattern of small openings that permit airflow while shielding the internal components from debris and direct impact. The lofted exterior geometry of the protective chamber further protects the sensor chamber from mechanical loads associated with collision.

The sensor chamber houses a pressure sensor, a microcontroller, a Bluetooth module, and a rechargeable battery. These electronics measure respiratory pressure during use and transmit data wirelessly to a companion application, providing the athlete with real-time and longitudinal feedback on respiratory muscle strength and breathing performance.

The device is designed to be manufactured at a target cost below \$30 and sold at a consumer price point of approximately \$100. The mouthguard and resistance chamber are produced as separate modules assembled with adhesive, allowing each subsystem to be optimized for its material and functional requirements independently.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a wearable respiratory training device according to one embodiment of the invention.

FIG. 2 is a vertical cross-sectional view of the respiratory training device of FIG. 1, showing the internal airflow path, the threshold flap, and the insides of the protective and sensor chambers.

FIG. 3 is a horizontal cross-sectional view of the respiratory training device of FIG. 1, specifically highlighting the resistance wall and its movement limits in its slot.

DETAILED DESCRIPTION OF THE INVENTION

Referring to FIG. 1, a respiratory training device **10** according to one embodiment of the invention comprises a mouth guard **4**, an airway inlet **3**, a sensor chamber **2**, and a protective chamber **1**. The entire device **10** is preferably formed of an injection molded silicone to maintain a compliant structure that is collision resistant. Other embodiments of this invention could form the mouthguard or other sections of the device of other compliant injection-molded or 3-D printed materials such as TPU.

As shown in the cross-sectional view of FIG. 2, the threshold flap **7** occupies the boundary between the protective chamber **1** and the sensor chamber **2**. The threshold flap **7** thickness is dependent on the material, but should deform at an angle of 90° when the average athlete exerts 25 cm H₂O or 2.45 kPa, 20% of the average athlete's Maximum Inspiratory Pressure. In this embodiment, the threshold flap **7**

requires the athlete to breathe in and out with a minimum pressure, constantly challenging their respiratory muscles.

The structure of the protective chamber **1** is essential to protection of the threshold flap **7** and the electronics located in the sensor chamber **2**. In a preferred embodiment, the protective chamber **1** has a pattern of small holes **6** to protect the threshold flap **7** from any debris. Additionally, the protective chamber **1** should have a lofted exterior to protect the sensor chamber **2** from any direct impact.

As shown in the cross-sectional view of FIG. 3, inside the sensor chamber **2** is where the threshold wall **9** lies. The threshold wall **9** is controlled by using the resistance adjustment wheel **5**, moving along the threshold wall slot **8**. As the location of the threshold wall **9** is moved using the resistance adjustment wheel **5**, the wall covers more or less of the airway inlet **3**, providing more or less resistance to athletes in which they can manual control.

In other embodiments of the invention, the protective chamber **1** and the sensor chamber **2** can be simplified and smaller in the case that app connected feedback is not present. Additionally, in other embodiments the threshold flap **7** and the resistance wall **9** can be replaced with other resistance providing mechanisms as long as they can be protected with compliant and collision resistant materials.

ABSTRACT

A compact and collision resistant mouth-held respiratory muscle trainer consists of a mouthguard, protective chamber, and sensor chamber. The sensor chamber includes a pressure sensor with a bluetooth module and a rechargeable battery allowing for app-connected feedback on the athletes respiratory progression. The resistance is applied through a threshold flap which deforms allowing airflow when the athlete breathes using the required threshold pressure, and a threshold wall which can be manually controlled to cover more of the airway inlet increasing respiratory resistance. The structure of the device allows for this invention to be used during contact sports and regular training for all athletes.

INVENTOR DECLARATION AND SIGNATURE

I hereby declare that I am the original inventor of the subject matter disclosed in this provisional patent application.

Name: Michael A. Alexiadis Date: 05/15/2026

Michael A. Alexiadis, Inventor

Executive Summary:

1. What are the desired function(s) of your design?

The Breathlete provides manually adjustable resistance to an athlete's respiratory muscles during normal training to build respiratory strength and improve cardiovascular endurance. Resistance is delivered through a threshold flap requiring a minimum pressure to open on every breath and a scroll wheel-controlled resistance wall that varies airflow restriction. A secondary function is measuring and transmitting breathing performance data to a companion app via an integrated pressure sensor and Bluetooth module.

2. What constraints related to the main function(s) must your design satisfy?

The device must be mouth-held to keep both hands free during athletic activity, fabricated from compliant silicone so collisions in contact sports do not injure the wearer or opponents, and allow resistance to be manually adjusted during use without removing the device. The protective chamber must shield internal electronics from debris and impact. The device must remain under 100 g and be manufacturable at a cost supporting a \$100 retail price.

3. What are the performance objectives of your design? (Give quantitative metrics as much as possible).

Retail price: \$100; BOM-verified unit cost: \$20.92 (~\$28–31 with assembly)

Total device mass: ~16.9 g (BOM-verified), target <100 g

Resistance valve range: 5–40 cmH₂O

Threshold flap deflection: ~25 cmH₂O (2.45 kPa), ~20% of average athlete Maximum Inspiratory Pressure

Resistance flexibility: 6–10 discrete levels

Mouthpiece hardness: Shore A 40–60

Pressure sensor accuracy: <±5%

Mouthpiece wall thickness: 10–15 mm

Factor of safety above 1 for all structural components under athletic loading

4. What alternative design concepts were considered?

Three historical product categories were evaluated: handheld inspiratory muscle trainers (e.g., POWERbreathe Plus), restrictive training masks (e.g., Elevation Training Mask 2.0), and app-connected mouth-held devices (e.g., Airofit Pro 2.0). All three scored 1 out of 5 on improvement feedback in the competitor HOQ. POWERbreathe and Airofit scored 1 on portability during training. The Elevation Mask scored poorly on comfort and adjustability. None satisfied contact-sport safety, hands-free use, adjustable resistance, and app feedback simultaneously.

5. What analyses were used to select among these alternative design concepts?

A conjoint analysis (n=10) showed strong customer preference for manually adjustable resistance, app-connected feedback, and a \$100 price point, eliminating fixed-resistance and feedback-free concepts. Competitor HOQ benchmarking confirmed no existing product adequately addressed portability, adjustability, feedback, and contact-sport safety together, validating the mouth-held compliant design as the only concept satisfying all primary constraints. TRIZ contradiction analysis was also applied to resolve key design tensions including resistance effectiveness vs. comfort and feature richness vs. cost.

6. What industry or society standards were used to inform or evaluate your design?

ISO 10993 guided biocompatible material selection for the mouthpiece given direct oral contact during use. RoHS and WEEE directed electronics material choices and end-of-life planning for the sensor, BLE module, and battery. UL2054 informed battery safety requirements for the rechargeable lithium cell. FCC Part 15 governs the BLE wireless module for U.S. market compliance. UN38.3 applies to lithium battery shipping and handling.

7. Which concepts or skills learned in your coursework were applied to the design? Projects are expected to make substantial use of MAE and related ENGRD classes. Please provide a list with each entry providing the department and number of the course, plus a brief description of the particular concept or skill used.

MAE 4631 (Advanced Product Design): Applied the full product development process including two-level HOQ, conjoint analysis, TRIZ and substance-field analysis, functional decomposition, FMEA, requirements traceability matrix, BOM development, and manufacturing process selection.

MAE 3230 (Fluid Mechanics): Used pressure-flow relationships to calculate the aerodynamic load on the threshold flap, determining the geometry needed to achieve the 25 cmH₂O deflection target.

MAE 3270 (Mechanics of Materials): Performed static analysis of the sensor chamber walls under worst-case collision loading and cyclic fatigue analysis of the threshold flap to produce a fatigue factor of safety.

MAE 2250 (Mechanical Synthesis): Applied the mechanical design process to develop and iterate on the device's physical design.

8. Evaluate your design, relative to its function(s) and constraints. How well did your design meet each of the performance objectives? How well does your design compare to other, existing solutions to the problem?

The design satisfies all primary functional and constraint requirements. The mouth-held silicone form factor addresses the contact-sport safety constraint that disqualifies every benchmarked competitor. The dual-mechanism resistance system and integrated electronics offer capabilities no existing product provides simultaneously. All quantitative performance objectives were met

through analysis: BOM-verified unit cost of \$20.92, device mass of ~16.9 g, threshold flap sized to 25 cmH₂O, wall FS > 2 under collision loading, and flap fatigue FS of 2.49. A physical prototype has not yet been built, as the course focused on the design and analysis phase; some material and thickness adjustments would likely follow prototype testing.

9. What impact do you see your design, if implemented, having upon public health, safety, and human welfare, as well as upon current global, cultural, social, environmental, and economic concerns?

The Breathlete is targeted at high school, collegiate, and professional athletes, a population that is generally already healthy, limiting its broad public health impact. Within that audience, the device lowers the barrier to respiratory muscle training by integrating it into normal practice without dedicated extra time, supporting improved cardiovascular endurance and breathing performance. The compliant silicone construction also allows it to double as a mouthguard. At \$100 in a U.S. athletic goods market valued at \$37.6 billion, it is financially accessible relative to comparable products. End-of-life considerations are addressed through modular disassembly and WEEE-aligned electronics recycling.

10. What format did your design take? For example, is it a complete set of CAD drawings, a working prototype, a full finished product, a system configuration, a process map, or something else?

The design was produced as a complete set of CAD drawings and renders, supported by fluid mechanics calculations, static and cyclic structural analysis, a full sourced BOM, injection molding cost analysis, and a provisional patent application. A fully working prototype was not developed, as the course focused on the design and analysis phase rather than physical production.

11. Describe each student's role in the design project if it was a group project.

Christopher Sawicki:

- Problem definition, basic functions, and special features of the Breathlete (1.3–1.5)
- Highest-level CAD and physical renderings (1.6)
- Full conjoint analysis across both test iterations and design decision impact (2.3)
- Geometric layout and module interface documentation (2.5.3–2.5.5)
- Mechanical analysis: static structural analysis and cyclic fatigue analysis of the threshold flap (2.11)
- Competitor HOQ Level 1 and Level 2 including BOM development (3.3)
- Full FMEA: failure mode identification, RPN calculation, and mitigation strategies (5.6)
- Provisional patent application: description of drawings and invention, abstract

Christopher Miranda Soto:

- HOQ Level 1 and full customer requirements list from empathy fieldwork and interviews (2.1–2.2)
- HOQ Level 2: EC rows, PC columns, correlation values, performance targets, and roof matrix (2.6)
- Target market identification and market sizing (3.1)
- Consumer behavior research and historical solutions (3.2)
- Bass Model adoption forecast (3.4)
- Full financial analysis: pricing, expected sales, NPV, COGS, and capital investment strategy (4.1, 4.4)

Michael Alexiadis:

- Product architecture: functional decomposition, clustered module schematic, and modular architecture justification (2.5.1–2.5.2, 2.5.5)
- TRIZ and substance-field analysis (2.8)
- Requirements traceability matrix including design gaps and scope creep (2.7, 2.10)
- Production cost analysis: manufacturing process selection, tooling parameters, CustomPartNet estimates (4.3)
- Legal requirements: safety regulations, liability, industry standards, and certifications (5.1–5.5, 5.7)
- Intellectual property considerations: patent, trade secret, and trademark strategy (5.8–5.9)
- Provisional patent application: field, background, summary